

MOVEMENTS OF SMALL INTESTINE

Headings

- ▼ MOVEMENTS OF SMALL INTESTINE
 - TYPES OF MOVEMENTS OF SMALL INTESTINE
 - ▼ 1. MIXING MOVEMENTS
 - A. Segmentation Contractions
 - B. Pendular Movements
 - ▼ 2. PROPULSIVE MOVEMENTS
 - A. Peristaltic Movements
 - Starling's Law of the Intestine
 - B. Peristaltic Rush (Rush Waves)
 - ▼ 3. PERISTALSIS IN FASTING (MIGRATING MOTOR COMPLEX - MMC)
 - Significance:
 - ▼ 4. MOVEMENTS OF VILLI
 - CONTROL OF VILLI MOVEMENTS

- Movements are essential for **mixing chyme with digestive juices, propulsion of food, and absorption.**

TYPES OF MOVEMENTS OF SMALL INTESTINE

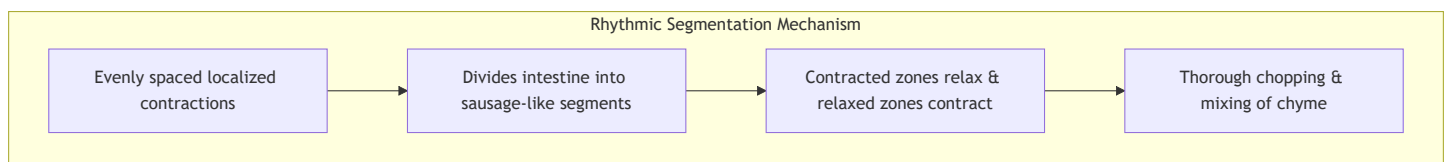
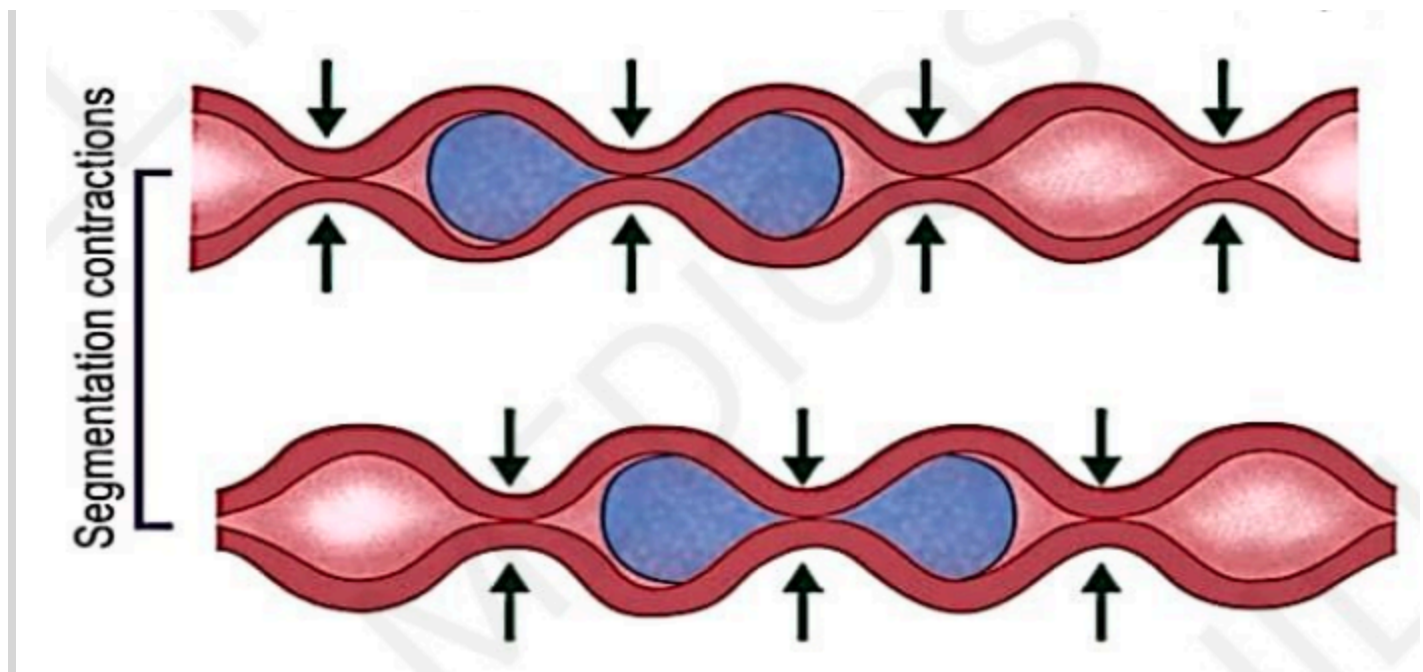
- a. **Mixing Movements :**
 - Segmentation movements
 - Pendular movements
- b. **Propulsive Movements :**
 - Peristaltic movements
 - Peristaltic rush
- c. **Peristalsis in Fasting :** Migrating Motor Complex (MMC)
- d. **Movements of Villi**

1. MIXING MOVEMENTS

Responsible for proper mixing of chyme with digestive juices such as **pancreatic juice, bile, and intestinal juice (succus entericus)**.

A. Segmentation Contractions

- Most common type of movement of the small intestine.
- Occur regularly or irregularly, but in a **rhythmic fashion** (*hence called rhythmic segmentation contractions*).
- Contractions occur at regularly spaced intervals along a section of the intestine.
- Segments of intestine between contracted segments remain **relaxed**.
- Alternate segments of contraction and relaxation give the **appearance of rings resembling a chain of sausages**.
- After some time, the contracted segments relax while relaxed segments contract.
- **Function** : Chops chyme many times, facilitating thorough mixing with digestive secretions.



B. Pendular Movements

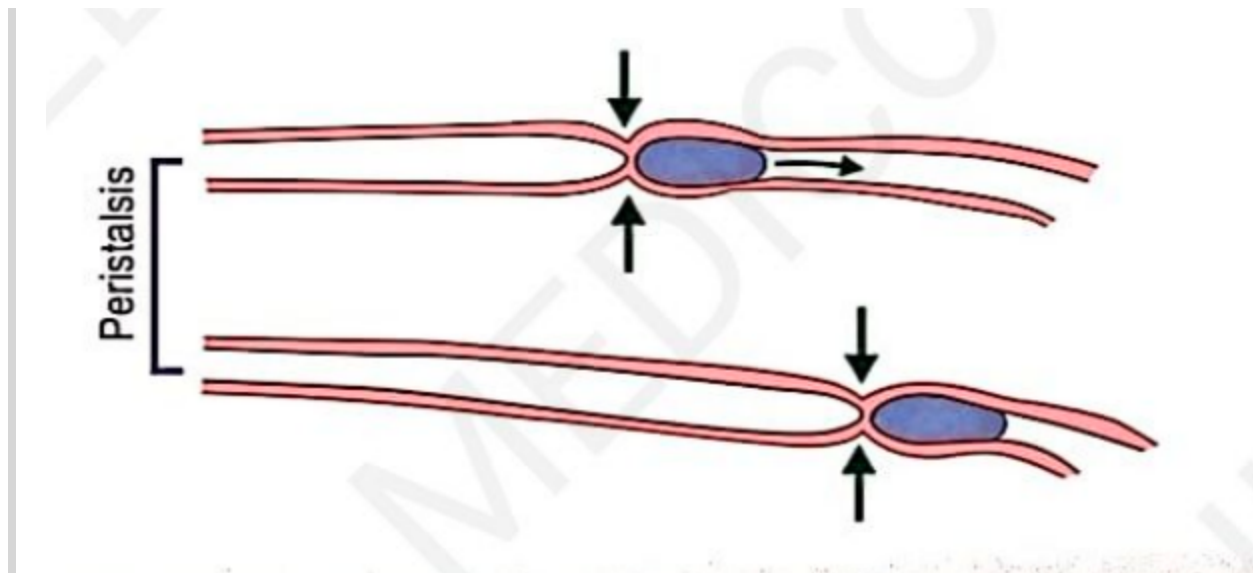
- **Sweeping to-and-fro movements** of the small intestine resembling the movement of a **clock pendulum**.
- Small portions (loops) of the intestine sweep forward & backward or upward & downward.
- **Function** : Helps in thorough mixing of chyme with digestive juices.

2. PROPULSIVE MOVEMENTS

Movements of the small intestine that push chyme in an **aboral direction** (towards the anus) through the intestinal tract.

A. Peristaltic Movements

- A wave of contraction followed by a wave of relaxation of muscle fibers.
- **Direction** : Always travels in an **aboral direction**.
- **Initiation** : Stimulation of smooth muscles of intestine initiates peristalsis.
- Although impulses travel from the point of stimulation in both directions, progress in an oral direction is rapidly inhibited; **only contractions in the aboral direction persist**.



Contraction Ring
(Upstream)

Pushes Bolus Aborally →

Relaxation Wave
(Downstream)

Starling's Law of the Intestine

- Formulated by **Ernest Starling** based on the direction of peristalsis.
- **Law Definition** : The response of the intestine to any local stimulus consists of a **contraction above (proximal to)** and a **relaxation below (distal to)** the stimulated area.
- **Characteristics of Waves** :
 - Normal peristaltic contractions are weak and disappear after traveling a few centimeters.
 - Chyme requires several hours to travel from the duodenum to the end of the small intestine.
 - Peristaltic waves increase significantly after a meal due to the **gastroenteric reflex** (*initiated by distension of the stomach, transmitted via myenteric plexus*).

B. Peristaltic Rush (Rush Waves)

- Powerful, sweeping peristaltic contractions caused by **excessive irritation of the intestinal mucosa**.
- Begins in the duodenum and passes rapidly through the entire small intestine, reaching the **ileocecal valve within a few minutes**.
- **Function** : Sweeps the contents of the intestine into the colon, relieving the small intestine of harmful irritants or excessive distension.

3. PERISTALSIS IN FASTING (MIGRATING MOTOR COMPLEX - MMC)

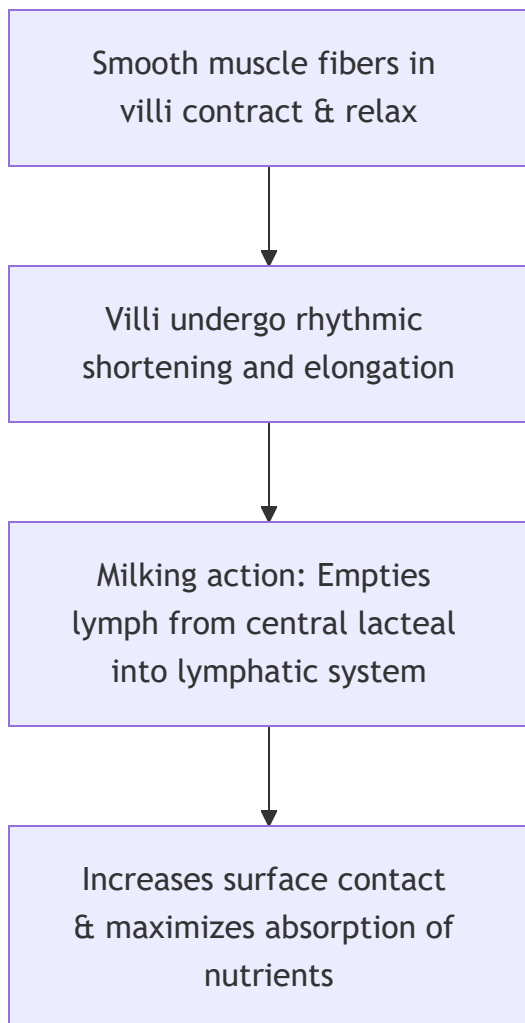
- Type of peristaltic contraction occurring in the **stomach and small intestine** during prolonged periods of **fasting (interdigestive state)**.
- Also called **Migrating Myoelectric Complex**.
- **Pattern & Rhythm** :
 - Involves large segments of the stomach and intestine (extends 20 – 30 cm).
 - Occurs cyclically once every 1.5 – 2 hours.
 - Starts as moderately active peristalsis in the body of the stomach and takes about **10 minutes to reach the colon**.

Significance:

- Sweeps excess digestive secretions into the colon, preventing accumulation of secretions in the stomach and intestine.
- Sweeps residual, undigested material into the colon (*acts as an intestinal housekeeper*).

4. MOVEMENTS OF VILLI

- Intestinal villi display individual movements simultaneously with overall intestinal motility.
- Occurs due to the **extension of smooth muscle fibers from the intestinal wall into the core of each villus**.



CONTROL OF VILLI MOVEMENTS

- a. **Neural Control** : Caused and regulated by local enteric nervous reflexes.
- b. **Hormonal Control** : Regulated by the hormone **Villikinin**, secreted from the small intestinal mucosa to stimulate villus motility.